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Seizure detection using the wristband accelerometer, gyroscope, and surface electromyogram signals based on in-hospital and out-of-hospital dataset

Guangming Wang [a,1], Hao Yan [b,1], Wen Li [a], Duozheng Sheng [a], Liankun Ren [c], Qun Wang [d,e], Hua Zhang [f], Guojun Zhang [g], Tao Yu [b,*], Gang Wang [a,**]

[a] The Key Laboratory of Biomedical Information Engineering of Ministry of Education, Institute of Biomedical Engineering, School of Life Science and Technology, Xi'an Jiaotong University, Xi'an 710049, PR China
[b] Department of Functional Neurosurgery, Beijing Institute of Functional Neurosurgery, Xuanwu Hospital, Capital Medical University, Beijing 100053, PR China
[c] Department of Neurology, Xuanwu Hospital, Capital Medical University, Beijing 100053, PR China
[d] Department of Neurology, Beijing Tiantan Hospital, Capital Medical University, Beijing 100070, PR China
[e] China National Clinical Research Center for Neurological Diseases, Beijing 100070, PR China
[f] Department of Neurosurgery, First Affiliated Hospital, Xi'an Jiaotong University, Xi'an 710061, PR China
[g] Department of Functional Neurosurgery, National Center for Children's Health, Beijing Children's Hospital, Capital Medical University, Beijing 100045, PR China

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ABSTRACT

Objective: Wearable devices are effective for detecting generalized tonic-clonic seizures (GTCS). However, many daily activities are often misclassified as GTCS, leading to a decline in user confidence. This study recommends utilizing wristband three-axis accelerometer (ACC), three-axis gyroscope (GYRO), and surface electromyography (sEMG) signals for GTCS detection and presents a novel seizure detection algorithm that offers high sensitivity and a reduced false alarm rate (FAR).

Methods: Inpatients with epilepsy and out-of-hospital healthy subjects were recruited and required to wear a wristband device to collect wristband signals. The proposed algorithm comprises five steps: preprocessing, motion filtering, feature extraction, classification, and postprocessing. The variations in performance across different signal combinations were compared. Additionally, the impact of training the model using only inpatient data versus the complete dataset on the algorithm's performance was also investigated.

Results: Wristband signals were collected from 45 patients and 30 healthy subjects, encompassing a total of 3367.3 h and including 60 GTCS. The proposed algorithm achieved 100 % sensitivity and a FAR of 0.1070/24 h. It demonstrated higher sensitivity and lower FAR compared to combinations with fewer signal modalities. In addition, the model trained on only in-hospital data demonstrates high sensitivity (98.33 %) and high FAR (0.9845/24 h).

Significance: The algorithm proposed for detecting GTCS using wristband ACC, GYRO, and sEMG signals achieved encouraging results, demonstrating the feasibility of this signal combination. Furthermore, incorporating out-of-hospital data into model training proved to be an effective solution for reducing FAR, which could facilitate the clinical application of seizure detection algorithms.

1. Introduction

Epilepsy is a chronic neurological disorder affecting approximately 70 million people worldwide [1]. One-third of patients experiencing recurrent seizures despite appropriate pharmacological management [2, 3]. Among these individuals with chronic refractory epilepsy, sudden unexpected death in epilepsy (SUDEP) remains the most prevalent cause of mortality, with an incidence rate of approximately three per

* Corresponding author.
** Corresponding author.
E-mail addresses: yutaoly@sina.com (T. Yu), ggwang@xjtu.edu.cn (G. Wang).
[1] Guangming Wang and Hao Yan contributed equally to this study.

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thousand, accounting for 10–50 % of all epilepsy-related deaths [4,5]. Generalized tonic-clonic seizures (GTCS) constitute one of the primary risk factors for SUDEP [6]. A recently published clinical practice guideline of the International League Against Epilepsy and the International Federation of Clinical Neurophysiology (IFCN) has provided evidence regarding the reliability of wearable devices for detecting GTCS including focal to bilateral tonic-clonic seizures (FBTCS) [7]. Thus, identifying biomarkers for GTCS and developing accurate real-time detection algorithms is crucial for enhancing patient safety through timely alerts and care.

Wearable devices for seizure detection commonly use physiological signals like three-axis accelerometer (ACC) [8–12], three-axis gyroscope (GYRO) [13–15], surface electromyography (sEMG) [16,17], photoplethysmography (PPG) [18,19], and electrodermal activity (EDA) [20,21]. Among these, ACC, GYRO, and sEMG exhibit strong correlations with the clinical manifestations of GTCS, leading to their extensive use in GTCS detection. Wearable devices that can simultaneously capture ACC, GYRO, and sEMG signals mainly include wristband, armband, and patch devices. Given that patients prioritize not only performance but also the aesthetics and comfort of seizure detection devices [22], wristband devices, being the most widely accepted by consumers, were selected as the data collection device for this study. Prior studies employing sEMG and other signals for seizure detection have exclusively utilized arm-mounted EMG sensors, whereas the feasibility of detecting seizures through multimodal fusion of wristband ACC, GYRO, and sEMG signals has not been systematically validated. We hypothesize that multimodal fusion of wristband sEMG with ACC and GYRO signals could enable algorithms to perceive seizure events from complementary multiple perspectives, thereby enhancing detection performance. This study aims to experimentally validate this hypothesis.

Many researchers have employed wearable devices to detect GTCS in inpatients, achieving sensitivities of 90–100% and 0.2–2/24 h false alarm rates (FAR) [9–12,16,21,23]. High FAR, which can diminish the benefits of detection, remains a significant limitation in the clinical application of current seizure detection algorithms [7]. Data from hospitalized patients may not accurately reflect real-world conditions due to the patients' restricted activity, leading to a high FAR. In contrast, outpatient populations, characterized by unrestricted mobility, provide data that more accurately reflects real-world distributions. However, the absence of a gold standard for seizure events in outpatient data poses challenges for model training [24]. Given that patients' behaviors during non-seizure intervals resemble those of healthy individuals, we recommend collecting data from healthy subjects outside clinical settings to enrich non-seizure data and improve the overall quality of the seizure dataset.

This study presents a novel seizure detection algorithm based on wristband ACC, GYRO, and sEMG signals. Firstly, multimodal data from the wristband devices, both inside and outside the hospital, underwent segmentation and preprocessing. Subsequently, suspected seizure samples were selected using motion filtering, followed by feature extraction and classification employing a support vector machine (SVM). Finally, the classification results were subjected to postprocessing to identify seizure events. The performance of the algorithm was rigorously evaluated using five-fold cross-validation, confirming the effectiveness of the multimodal wristband signals employed in seizure detection. Additionally, the findings demonstrated the role of out-of-hospital data in reducing FAR.

2. Materials and methods

2.1. Participants

Patients with epilepsy were recruited from four clinical sites: Xuanwu Hospital, Beijing Tiantan Hospital, Beijing Children's Hospital, and the First Affiliated Hospital of Xi'an Jiaotong University. The study was conducted from March 2022 to September 2023 with collected clinical data. The inclusion criteria for patients were defined as follows: 1) age between 5 and 60 years; 2) a history of GTCS or FBTCS; and 3) more than one seizure per month over the preceding 3-month period, with frequency data verified through patient self-reports. Healthy participants meeting the following inclusion criteria were subsequently recruited: (1) age range 5–60 years; (2) absence of a history of epileptic seizures; and (3) demonstrated willingness to wear the wrist throughout the study protocol. The study was approved by the institutional review board of four hospitals (LYS[2022]012), and all the participants or their caregivers signed written informed consents.

2.2. Data collection

During monitoring, patients were equipped with both an electroencephalogram (EEG) monitor and a wristband device to collect video-EEG (v-EEG) data alongside wrist-based ACC, GYRO, and sEMG signals. The v-EEG signals were recorded using the EEG-1100 Video-EEG system produced by NIHON KOHDEN Company. 19 electrodes were placed according to the International 10–20 system, with a sampling rate of 200 Hz. The Cz electrode was utilized as the reference electrode. A band-pass filter with a frequency range of 0.5 Hz to 60 Hz was applied to eliminate low-frequency artifacts and high-frequency noise from the EEG data. The wristband device, developed by our laboratory, employs the BMI160 6-axis inertial sensor from Bosch Sensortec to capture ACC and GYRO signals (each at 50 Hz), and utilizes the GH3220 health sensor from Goodix to collect sEMG signal at 200 Hz. To minimize motion artifacts, patients were instructed to correctly wear the wristband device to prevent noticeable displacement during rapid movements. Inpatients with asymmetrical seizure symptoms wore the wristband device on the side with more severe symptoms, while the remaining patients and healthy participants wore it on their non-dominant side. Additionally, temporal offsets between EEG and wristband devices were documented prior to each data acquisition session. Following temporal synchronization using these offsets, the wrist-worn device data were annotated with seizure onset/offset timestamps derived from gold-standard EEG recordings. Fig. 1 provides a schematic representation of the EEG and multimodal wristband signals recorded during a GTCS event, highlighting specific variations in wristband signals associated with seizure activity.

2.3. Reference standard

Three experienced clinicians independently reviewed the v-EEG data to determine seizure characteristics, including seizure type, onset time, and offset time. Seizure type was determined according to the criteria established by the International League Against Epilepsy (ILAE) in 2017 [25]. In cases of disagreement among the clinicians, majority rule was used to get the only final marker. GTCS and FBTCS were categorized as target (positive events), and other seizure types were regarded as non-seizures (negative events).

2.4. Seizure detection algorithm

Fig. 2 illustrates the overall workflow of our algorithm, along with the data partitioning methodology employed. The algorithm was proposed based on wristband ACC, GYRO and sEMG signals, consisting of five steps: preprocessing, motion filtering, feature extraction, classification, and postprocessing. The algorithm's detection performance is evaluated using five-fold cross-validation, with model training conducted separately on both the entire dataset and the in-hospital dataset.

Preprocessing. First, the raw data were segmented into 6-second epochs, with a 5-second overlap between consecutive epochs. Second, each epoch underwent signal quality evaluation using a custom signal quality index. Epochs deemed of good quality were retained for subsequent analysis, while those classified as poor quality were excluded for postprocessing. Third, the three-axis ACC data were consolidated into a

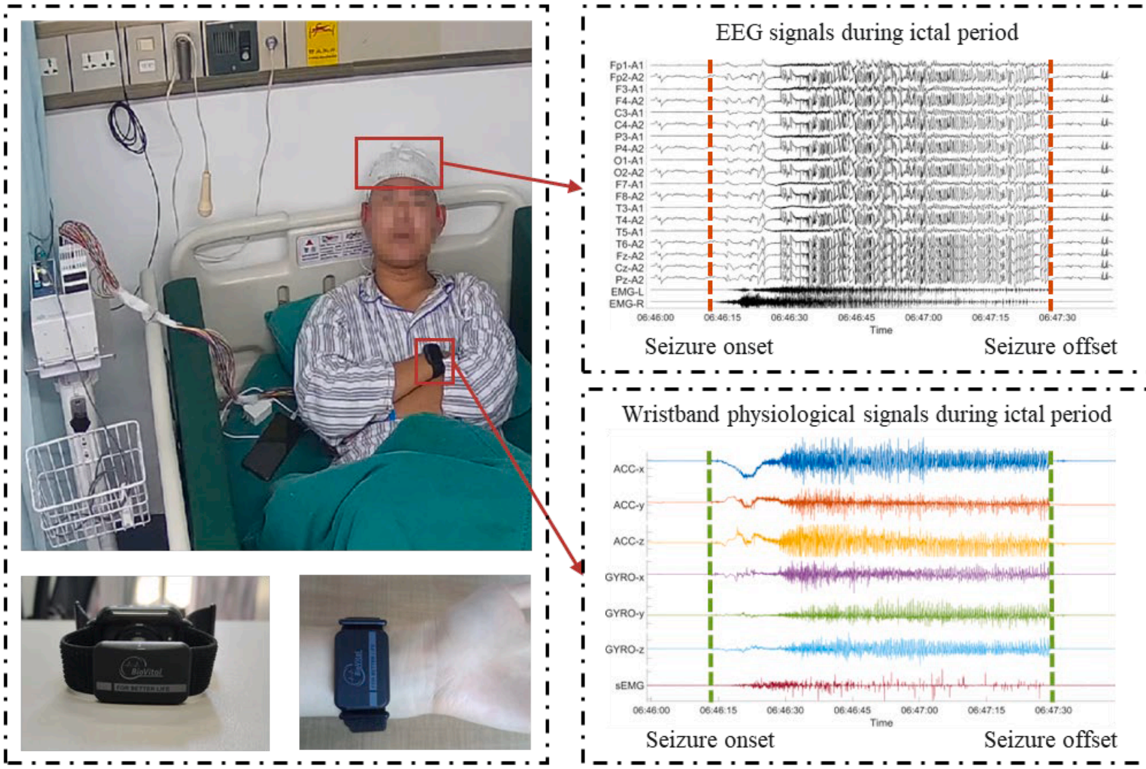

Fig. 1. The wristband device used in our work, and raw data of a representative seizure.

net ACC $\left(\mathrm{ACC}_{net} = \sqrt{\mathrm{ACC}_x{}^2 + \mathrm{ACC}_y{}^2 + \mathrm{ACC}_z{}^2}\right)$. Finally, given that over 99% of the ACC power was distributed below 15 Hz [26], the net ACC was processed through low-pass and high-pass filters with cut-off frequencies of 0.5 Hz and 15 Hz, respectively. The three-axis GYRO data were similarly processed to obtain the filtered net GYRO. sEMG signals were filtered using a 10 Hz high-pass filter and a 49–51 Hz notch filter to eliminate motion artifacts and power line interference. All filters were implemented as zero-phase Butterworth filters, ensuring an amplitude gain of 1 within the passband.

Motion filtering. GTCS are often accompanied by vigorous limb and body movements, while motion during most daily activities, especially sleep, is minimal. Thus, the mean of absolute net ACC (ABM), serving as a quantitative indicator of motion amplitude, was utilized for screening suspected seizure epochs, aiming to minimize data imbalance to the greatest extent possible. An epoch's ABM exceeding a predefined threshold indicates strong motion and potential seizure activity, whereas epochs with ABM below the threshold are classified as non-seizure states with a seizure occurrence probability of zero. The threshold was set at the 20th percentile of the ABM calculated across all seizure epochs.

Feature extraction. A total of 71 time-domain and frequency-domain features were extracted from the net ACC, net GYRO and sEMG data. The time-domain features were comprised of the mean absolute value, root mean square, standard deviation, quartile range, waveform length, energy, jerk, kurtosis, skewness, etc [23,27]. Frequency-domain features were derived from the power spectral density of the signals using the Welch method, which involved a 4-second rectangular window and 50% overlap, including relative power, percentile frequency, peak frequency, average frequency, median frequency, maximum power spectral density, etc [28,29].

Classification. The SVM distinguishes between two data classes by identifying a hyperplane that maximizes the margin, relying on a limited number of support vectors, which enhances its generalization performance [30]. Due to mild data imbalance after motion filtering, this study chose a linear SVM (SVM-L) to ensure stable seizure detection. The cost parameter was set to the default value of 1. The output of the SVM-L reflects the distance between the epochs and the hyperplane, which ranges over all real numbers. In this context, a sigmoid function was employed to transform this distance into the probability of seizure occurrence.

Postprocessing. Postprocessing involves three key steps: linear interpolation, mean filtering, and seizure event decision. The postprocessing phase receives input from preprocessing, motion filtering, and SVM-L. Epochs marked as poor quality in preprocessing were excluded from seizure probability calculations. To ensure a continuous probability curve, linear interpolation was employed to fill the gaps left by these poor-quality epochs. Following this, a forward mean filter was applied to mitigate high-frequency noise [31]. Lastly, a seizure event was determined based on a custom-defined criterion. Specifically, if the seizure probability exceeds 0.5 for a continuous period of 9 epochs, a seizure event is deemed to have occurred. Furthermore, any detected seizures within the subsequent 5 minutes are considered part of the same seizure event.

Evaluation criteria. To validate the efficacy of out-of-hospital data, the model training was conducted using both the entire dataset and solely in-hospital data. For testing, both in-hospital and out-of-hospital data were utilized, as depicted in Fig. 2B. In addition, A five-fold cross-validation procedure was employed to evaluate the proposed algorithm. The dataset was divided into five groups, each containing one-fifth of the data from patients and one-fifth from health controls, ensuring that no individual's data appears in more than one group. The evaluation metrics included sensitivity (SEN), FAR, detection latency (LAN), in-hospital FAR (FAR_1), out-of-hospital FAR (FAR_2), and the balance between sensitivity and FAR (SF). A seizure event detected by the model during ictal periods was classified as a true positive (TP). Conversely, a failure to detect a seizure during ictal periods was classified as a false negative (FN). The sum of TP and FN events corresponded to the total number of seizures. A seizure detected during interictal periods was categorized as a false alarm, or false positive (FP).

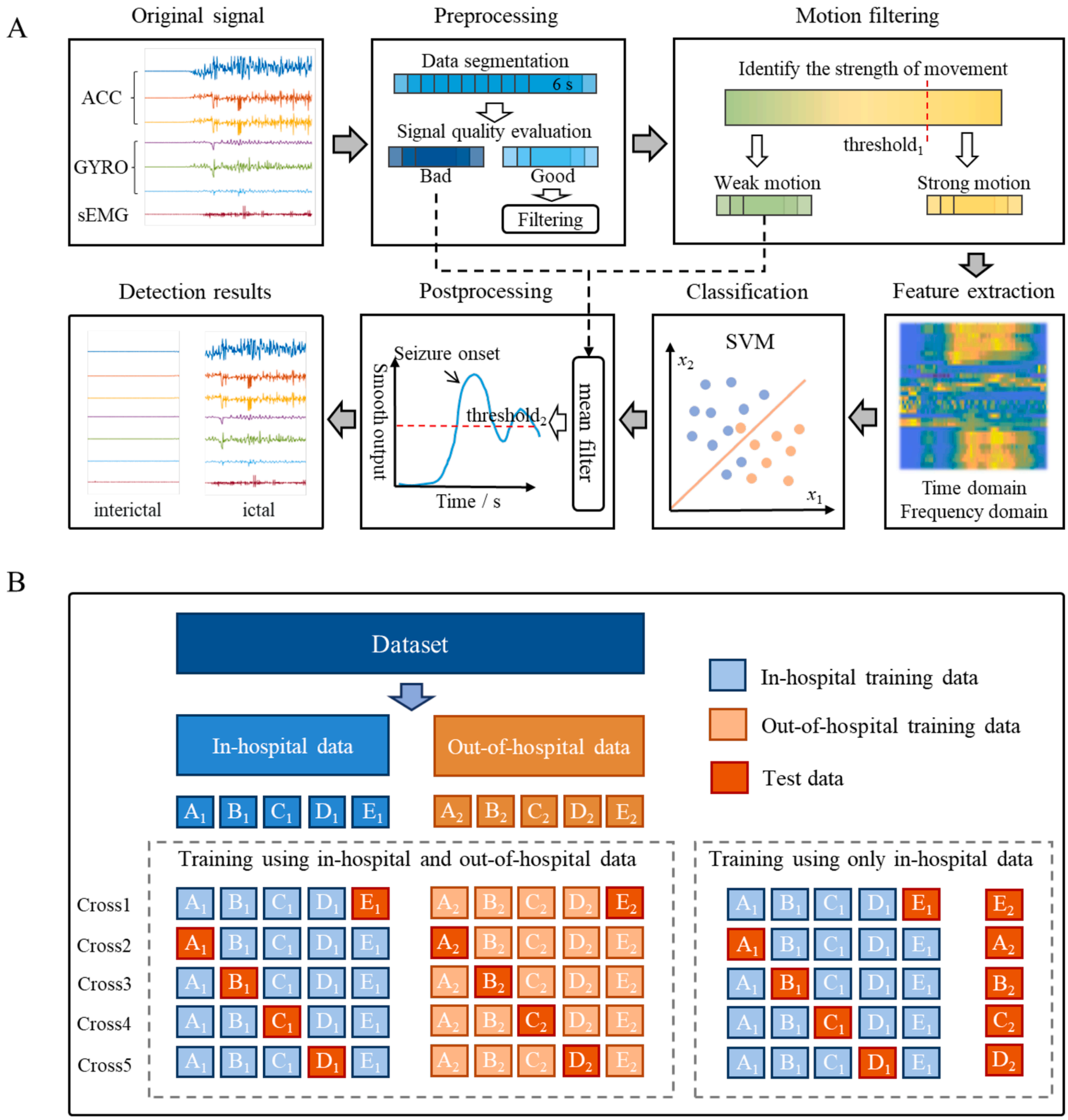

Fig. 2. Flow chart of our proposed seizure detection algorithm. (A) The data flow process of algorithms. (B) The process of algorithm dataset partitioning.

Sensitivity, FAR, and latency were identified as critical performance indicators [32]. Sensitivity was defined as the ratio of the number of TP to the total number of seizures. FAR was quantified as the number of false alarms per 24 hours. Latency was calculated as the interval between the model's detection of a seizure and the actual seizure onset. In addition, FAR_1 was defined as the false alarms per 24 hours within the in-hospital data, while FAR_2 was defined as the false alarms per 24 hours within the out-of-hospital data. Given the inherent difficulty in optimizing both sensitivity and FAR simultaneously, the index SF was employed to identify the optimal model [33]. The SF was computed as follows:

$$SF = \sqrt{\frac{SEN^2 + (1 - FAR')^2}{2}} \tag{1}$$

$$FAR' = \begin{cases} FAR, & FAR \le 1/24h \\ 1, & FAR > 1/24h \end{cases} \tag{2}$$

3. Results

A total of 45 patients (Xuanwu Hospital, n = 21; Beijing Tiantan Hospital, n = 16; Beijing Children's Hospital, n = 3; and the First Affiliated Hospital of Xi'an Jiaotong University, n = 5) and 30 healthy subjects were recruited, providing a cumulative recording duration of 3367.3 h. Patients contributed 1582.6 hours, while healthy subjects contributed 1784.7 h. During this period, 60 vEEG-confirmed seizures were documented, with durations of 41 to 152 sec; each patient experienced 1 to 3 valid seizures. Table 1 presents demographic and data information for all participants. Further, statistical analysis comparing age and recording duration between patient and healthy control groups were performed. Due to non-normal data distribution and heteroscedasticity, Mann-Whitney U test was selected for both comparisons [34]. For age analysis, the test revealed no significant difference between groups ($p = 0.471$), as their age distributions predominantly overlapped in the 20–40 year range despite a lower minimum age in patients. In contrast, recording duration analysis revealed a statistically significant difference ($p = 9.2 \times 10^{-5} < 0.001$), potentially undermining

Table 1
Demographic information and data information of patients and healthy subjects.

	In-hospital patients	Out-of-hospital healthy subjects	All subjects
Number of participants, n	45	30	75
Age, years (mean)	8–44 (23.9)	17–38 (25.2)	8–44
Female, n (%)	13 (29)	14 (47)	27
Recording duration, hour (mean)	1582.6 (35.2)	1784.7 (59.5)	3367.3
Number of seizures, n	60	0	60
Seizures per patient, n (mean)	1–3 (1.3)	——	0–3
Seizure duration, s (mean)	41–152 (62.0)	——	41–152

the study's reliability. This difference arose because healthy participants consistently provided ≥48 hours of valid data, whereas patients often could not meet this requirement due to early discharge or technical issues like loose sensors.

The performance of the algorithm was assessed using different combinations of modalities and different datasets, with the results presented in Table 2. For the model trained using the complete dataset, the best performance was achieved when utilizing signals from all modalities, yielding a sensitivity of 1, a FAR of 0.1070/24 h, an average latency of 29 seconds (range: 13–66 seconds), and a SF of 0.9480. In addition, all 15 false alarms occurred during daytime. The model exhibited the poorest performance when using only sEMG signals, with a sensitivity of 0.6333, a FAR of 0.1783/24 h, and an average latency of 25 s, demonstrating the feasibility of using wristband sEMG signals for seizure detection. Training with multimodal signal combinations resulted in enhancements across detection performance metrics compared to using each single signal, with sensitivities exceeding 0.9. Notably, while FAR1 for models using ACC matched that of all modal signals, FAR2 was significantly high, indicating that relying solely on ACC signals continues to present challenges in effectively differentiating between motor activity and seizure events. Furthermore, compared to training models using the entire dataset, when models were trained only using in-hospital data, they often exhibited higher sensitivity but also higher FAR, resulting in low SF. This could be because in-hospital data lacks daily activities, leading to seizure data showing more intense movements compared to non-seizure data. Consequently, the model identifies a hyperplane separating motion intensities rather than distinguishing seizures from high-intensity movements. The inclusion of out-of-hospital data can address this issue, highlighting the contribution of out-of-hospital data in enhancing the generalizability of the model.

The relevant performance metrics of the model using all signals for each subject, or each seizure event were recorded when employing signals from different modalities. Additionally, the variations in algorithm detection performance across various classical machine learning models were investigated. These results are depicted in Fig. 3. In terms of sensitivity, as illustrated in Fig. 3A and Table S1, the multimodal signal-based model can detect all seizures in the majority of patients, whereas the unimodal signal-based model tends to miss seizures in some patients. In addition, as shown in Fig. 3B and Table S2, the model utilizing multimodal signals demonstrated a FAR primarily within the range of 0–0.6/24 h, while the single-mode signal-based model exhibited a notable occurrence of high FAR (≥1.5/24 h) in several patients. Similarly, the latency of detected seizures in the multimodal signal-based model predominantly fell within the range of 10–40 s (Fig. 3C, Table S3, and Table S4). In contrast, the single-mode signal-based model often resulted in high latency detections (≥60 seconds). This indicates that the incorporation of multimodal signals not only enhances detection performance but also effectively mitigates the occurrence of extremely unfavorable scenarios. A comparative analysis of various classic machine learning models, including SVM-L, Adaboost, K-Nearest Neighbors (KNN), Artificial Neural Networks (ANN), Random Forest (RF), and SVM with a Gaussian kernel (SVM-RBF), was conducted. The results shown in Fig. 3D revealed that SVM-L exhibited optimal sensitivity and SF; however, it lagged behind Adaboost and RF regarding FAR and latency. The SVM-RBF model demonstrated the lowest performance overall, with the poorest sensitivity, and highest latency.

Performance metrics for each subject or seizure were assessed when training the models using either the entire dataset or the in-hospital data, and the results are presented in Fig. 4. As shown in Fig. 4A, the metrics for the in-hospital model were generally inferior to the all-data model, except for latency. In terms of sensitivity (Fig. 4B), both models exhibited similar performances, with the in-hospital model detecting one fewer event. However, as depicted in Fig. 4C, the FAR was notably higher for the in-hospital model when compared to the all-data model, and there were also more subjects with serious FAR. Regarding latency, Fig. 4D illustrates that the in-hospital model demonstrated superior performance, with the majority of seizure detection delays occurring within a 10–20 s interval, whereas the latency for the all-data model was predominantly concentrated in the 20–30 s range.

4. Discussion

A retrospective analysis was conducted on the original data and algorithm detection outcomes from 45 inpatients and 30 healthy subjects. The study found that wristband sEMG can be utilized for the detection of seizures. Utilizing a combination of multiple modalities for seizure detection, rather than relying solely on a single modality, results in improved sensitivity and reduced FAR. Furthermore, the inclusion of out-of-hospital data for model training was found to enhance the

Table 2
The results of the model based on different modality signals and training data.

Training data	Modes	Sens	FAR/24 h	SF	Latency, s	FAR_1	FAR_2
In-hospital data	ACC	0.9667	1.8191	0.6835	26±10	0.3644	3.1087
	GYRO	0.8667	1.0059	0.6128	31±12	0.3492	1.5880
	sEMG	0.6833	0.5422	0.5816	23±14	0.4403	0.6325
	ACC&GYRO	1	1.8548	0.7071	23±9	0.2733	3.2567
	ACC&sEMG	1	1.1200	0.7071	22±8	0.2885	1.8571
	GYRO&sEMG	0.9667	0.6991	0.7159	23±10	0.3644	0.9958
	ALL	0.9833	0.9845	0.6954	27±11	0.2126	1.6687
In-hospital and out-of-hospital data	ACC	0.8333	0.3068	0.7665	34±14	0.0304	0.5518
	GYRO	0.6833	0.1855	0.7518	34±14	0.0607	0.2961
	sEMG	0.6333	0.1783	0.7336	25±15	0.1366	0.2153
	ACC&GYRO	0.9333	0.2426	0.8500	33±12	0.0455	0.4172
	ACC&sEMG	0.9667	0.1427	0.9136	29±14	0.0607	0.2153
	GYRO&sEMG	0.9000	0.1712	0.8651	33±15	0.0759	0.2557
	ALL	1.0000	0.1070	0.9480	29±13	0.0304	0.1749

Abbreviations: Sens = sensitivity; FAR = false alarm rate in all the dataset; FAR_1 = false alarm rate in-hospital dataset; FAR_2 = false alarm rate in out-of-hospital dataset. The value of detection latency is the average of latency of all the detected seizures.

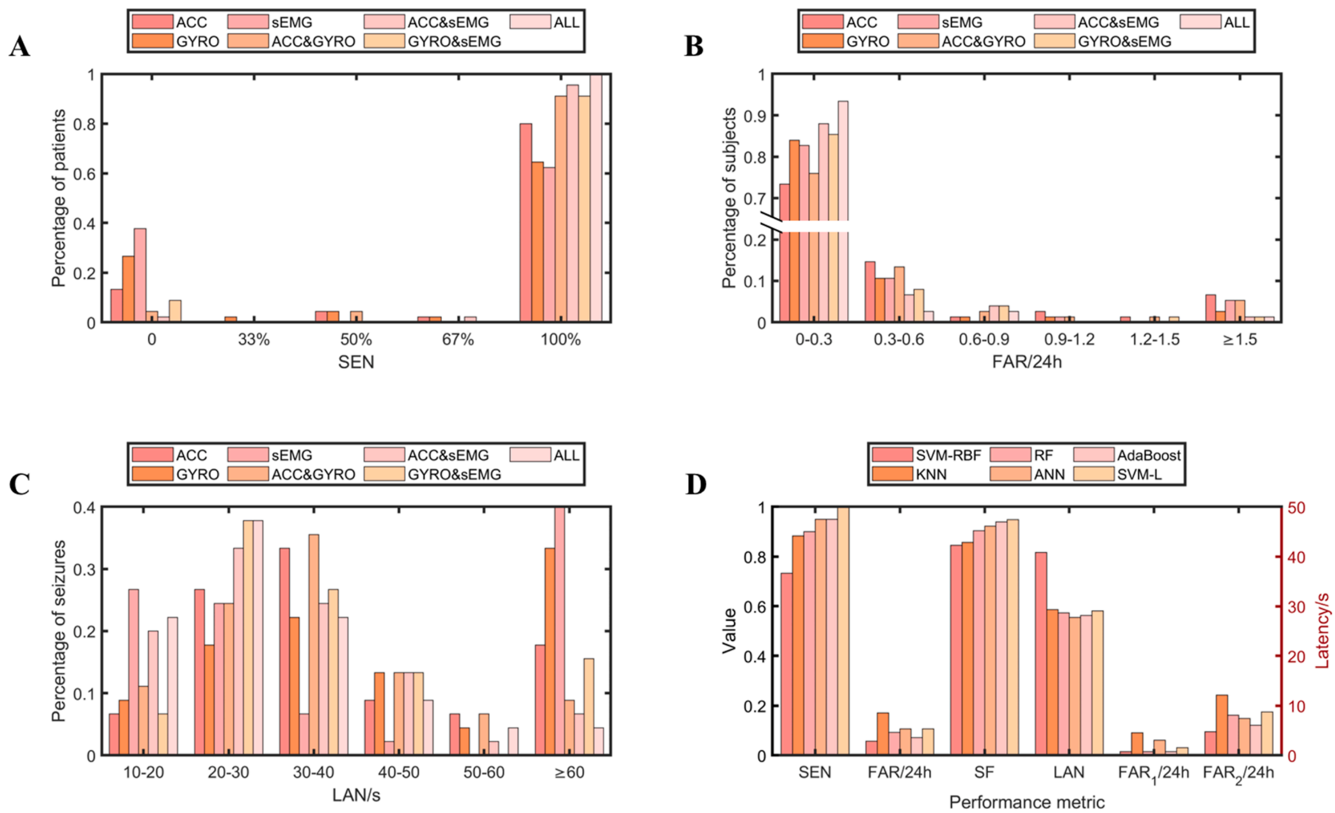

Fig. 3. The performance of algorithms when using different modal signals or machine learning classifiers. (A) Histogram depicting the sensitivity distribution across all patients. (B) Histogram illustrating the distribution of the false alarm rate for all subjects. (C) Histogram representing the latency of detection distribution for all identified seizure events. (D) Comparison of the SVM-L classifier with other traditional machine learning classifiers.

performance of the seizure detection algorithm.

4.1. The effect of wristband multimodal signals on seizure detection

Different signals can reflect specific aspects of clinical symptoms during epileptic seizures. For instance, both ACC and GYRO data primarily capture the periodic movements characteristic of the clonic phase in patients, while wrist-based sEMG signals effectively detect seizures by corresponding to muscle activity during the tonic and clonic phases [35]. The integration of diverse signal modalities can enrich the dataset with more relevant information pertaining to seizures, thereby enhancing the detection performance of the model. Analysis of Table S3 revealed that among 60 seizures, the ACC-based model missed 10, the GYRO-based model missed nineteen, and the sEMG-based model missed twenty-two. The ACC-based model and GYRO-based model together missed 8, while the model trained on both ACC and GYRO missed only 4. Similarly, the ACC-based model and sEMG-based model did not detect 2 together, and the model that combined ACC and sEMG also missed 2. Furthermore, the GYRO-based model and sEMG-based model jointly overlooked 4, whereas the model based on both GYRO and sEMG missed 6. When all three modalities were fused, the model detected all seizures successfully. This indicates that after the fusion of modalities, the model enables detection of seizures identifiable by any individual modality. This supports previous findings that multimodal integration outperforms single-modality detection [36,37], as its complementary features mitigate the limitations of any individual signal.

4.2. The effect of out-of-hospital data from healthy subjects on seizure detection

Many studies show that rhythmic activities are easily mistaken as seizures, such as exercise [38–40]. These activities are common at home, but not available in the hospital. As a result, the in-hospital dataset is too inadequate that the model can't learn the real data distribution and performs worse in out-of-hospital subjects than inpatients. Two studies support this. One team designed the Epi-care wristband device, which detects GTCSs by ACC. For inpatients, the sensitivity and FAR were 89.70 % and 0.2/24 h[9]. For outpatients, the sensitivity and FAR turned to 85 % and 1.4/24 h[38]. The other team invented an E4 wristband device using ACC and EDA to detect GTCSs. Similarly, the sensitivity for inpatients and outpatients was approximately equal (i.e. 94.55 % for inpatients and 93% for outpatients) [21], and the FAR was higher for outpatients (i.e. 0.2/24 h for inpatients and 0.58/24h for outpatients) [41]. Our findings further emphasize the necessity of incorporating out-of-hospital data to improve model generalization and real-world applicability

Seizure detection algorithms aim to work with wearable devices for home use, but they often perform better in hospitals than at home. Evaluating the model in hospital settings can lead to overly optimistic results. So, to get more reliable results, the model must be tested using data from out-of-hospital subjects. As for outpatients, visual review of video and self-reported by themselves [24,42,43] is necessary for relatively accurate labels, which need large time cost. Testing on healthy subjects, who cannot have seizures, is beneficial as it provides ample non-seizure data without the time costs associated with outpatients. During the development of seizure detection algorithm, it's a good solution to get accurate performance by using healthy subjects.

4.3. Limitations and future work

Data collection from multiple medical centers enhances the generalizability and reliability of the algorithm. However, the inconsistent progress in data collection across centers undermines this advantage. It is essential to continue the acquisition of additional epilepsy-related

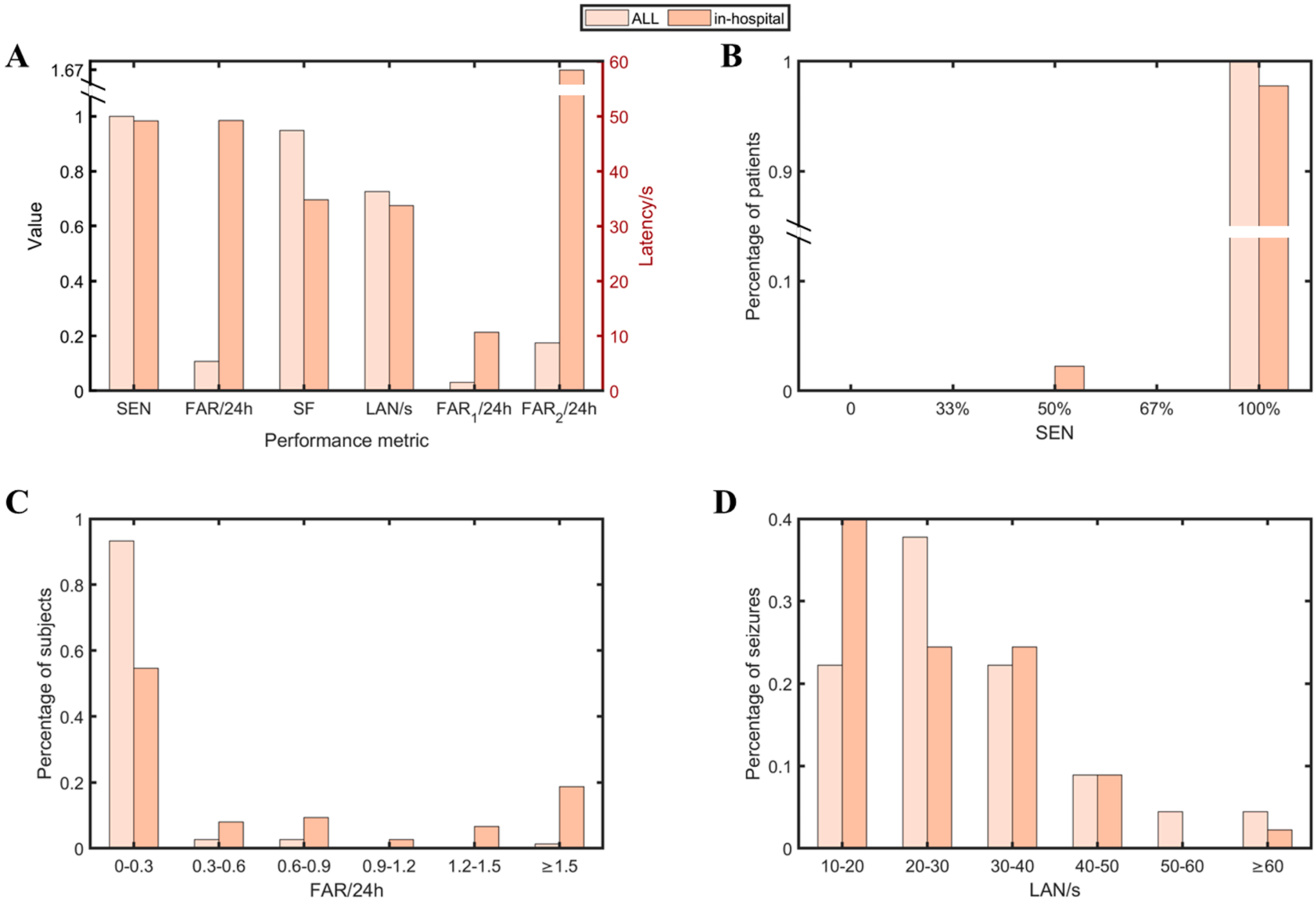

Fig. 4. Results of the models trained using both all data and in-hospital data. (A) Overall metrics for the two models. (B) Histogram depicting the sensitivity distribution across all patients. (C) Histogram illustrating the false alarm rate distribution for all subjects. (D) Histogram representing the latency distribution for the detection of all identified seizure events.

data to mitigate this issue. Then, our study utilized datasets with limited sample sizes and relatively short monitoring durations. Future validation will be conducted on larger-scale datasets to assess clinical effectiveness. Furthermore, this study used a threshold-based motion filtering approach to differentiate weak from strong motion, making the threshold selection critical. A common method involves setting the threshold empirically. In this experiment, the threshold was established at 0.14 g (g = 9.8 m/s^2) based on the net ACC magnitude during the seizure period. In contrast, two other studies set their thresholds to 0.1 g and 0.2 g, respectively [11,44]. This approach exhibits a considerable degree of subjectivity. An alternative method involves performing a grid search for the threshold using a validation set, which entails significant time costs. Future research could explore an adaptive learning model approach to optimally determine the threshold parameters. Additionally, the manual feature extraction in this study overlooks temporal information, potentially underutilizing seizure-related data. Models that extract temporal features should be explored to improve seizure detection algorithms. Concurrently, our algorithm is limited to detecting GTCS and FBTCS, excluding other seizure types. Expanding detectable seizure categories could enhance injury prevention in clinical practice. Lastly, the algorithm's real-world reliability remains limited by changing parameters and offline analysis. Future work should evaluate algorithm's performance with fixed parameters in epilepsy patients under real-world out-of-hospital conditions to enable clinical application.

5. Conclusions

This study proposed a novel seizure detection algorithm based on wristband ACC, GYRO, and sEMG signals. The algorithm was validated on a dataset comprising 45 inpatients and 30 healthy subjects, and achieved a sensitivity of 100 %, a FAR of 0.1070/24 h, a SF of 0.9480, and an average latency of 29 seconds. Our findings demonstrate that using wrist-based sEMG signals for detecting seizures is feasible. The combination of ACC, GYRO, and sEMG enhances algorithm performance beyond what is achievable with either two or a single modality. Additionally, training the model with out-of-hospital data recorded from healthy subjects significantly reduces the FAR and effectively improves the performance of the proposed algorithm.

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CRediT authorship contribution statement

Guangming Wang: Data curation, Conceptualization, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Hao Yan:** Data curation, Conceptualization, Investigation, Methodology, Validation. **Wen Li:** Conceptualization, Formal analysis, Methodology, Software, Writing – original draft. **Duozheng Sheng:** Data curation, Investigation, Methodology. **Liankun Ren:** Data curation, Validation. **Qun Wang:** Data curation, Validation. **Hua Zhang:**

Data curation, Validation. **Guojun Zhang:** Data curation, Validation. **Tao Yu:** Data curation, Project administration, Resources, Supervision, Writing – review & editing. **Gang Wang:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

None of the authors has any conflicts of interest to disclose. We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.seizure.2025.03.016.

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